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iT 邦幫忙

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簡易Git Server架設 - iT 邦幫忙

設定方式 · 現在vm中灌好一個ubuntu當作測試環境 · 在ubuntu中安裝好ssh(`sudo apt install ssh`) · 在ubuntu中建立一個資料夾，並在這個資料夾下執行`git init --bare`。



AI 摘要



+9

架設 Git 主機通常透過 SSH 協定來實現，最簡單的方法是在 Linux 伺服器上安裝 Git，建立一個專用的 git 使用者，並使用 `git init --bare` 建立裸儲存庫 (Bare Repository)。這能讓團隊成員透過 `git clone git@server:project.git` 進行存取。

iThelp +3

Linux (Ubuntu/Debian) 架設 Git Server

這是最標準的自架方式，具有權限管理功能。

1. 安裝 Git:

bash



```
sudo apt-get update
sudo apt-get install git-core
```

2. 建立 Git 使用者:

bash



```
sudo useradd -m git
sudo passwd git
```

3. 建立儲存庫 (裸庫):

切換到 git 使用者，在伺服器上建立一個沒有工作目錄的 `project.git` 資料夾。

bash



```
su - git
mkdir -p /home/git/repo/my-project.git
cd /home/git/repo/my-project.git
git init --bare
```

4. 客戶端連線:

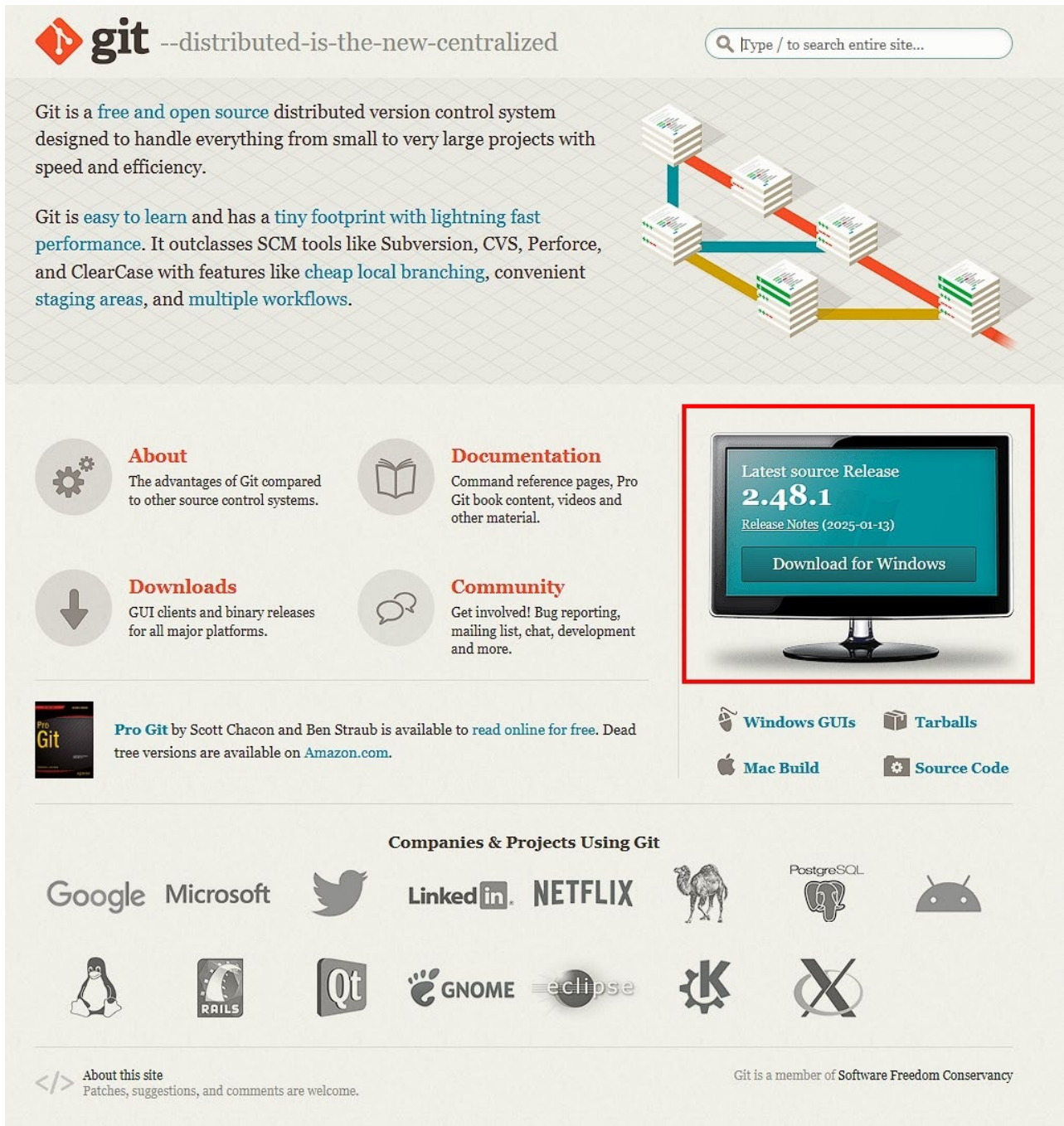
使用者可以使用 SSH key 或密碼進行 push/pull。

```
bash
```



```
git remote add origin git@<server_ip>:/home/git/repo/my-project.git
```

1. 安裝 Git 至 Git 官方網站下載並安裝 Git。



The screenshot shows the Git website homepage. At the top, the Git logo is followed by the tagline "--distributed-is-the-new-centralized". A search bar is on the right. The main content area describes Git as a free and open source distributed version control system, easy to learn, and having a tiny footprint with lightning fast performance. It lists features like cheap local branching, convenient staging areas, and multiple workflows. To the right is a diagram of a distributed version control system. Below this are four sections: About, Documentation, Downloads, and Community. A red box highlights the 'Latest source Release 2.48.1' section, which includes a 'Download for Windows' button. Below this are links for Windows GUIs, Tarballs, Mac Build, and Source Code. At the bottom, there's a section for 'Companies & Projects Using Git' with logos for Google, Microsoft, Twitter, LinkedIn, Netflix, PostgreSQL, Android, RAILS, Qt, GNOME, eclipse, K, and X. The footer includes 'About this site' and 'Git is a member of Software Freedom Conservancy'.

git --distributed-is-the-new-centralized

Git is a **free and open source** distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is **easy to learn** and has a **tiny footprint with lightning fast performance**. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like **cheap local branching**, convenient **staging areas**, and **multiple workflows**.

About
The advantages of Git compared to other source control systems.

Documentation
Command reference pages, Pro Git book content, videos and other material.

Downloads
GUI clients and binary releases for all major platforms.

Community
Get involved! Bug reporting, mailing list, chat, development and more.

Latest source Release
2.48.1
[Release Notes \(2025-01-13\)](#)
[Download for Windows](#)

Pro Git by Scott Chacon and Ben Straub is available to [read online for free](#). Dead tree versions are available on [Amazon.com](#).

[Windows GUIs](#) [Tarballs](#)
[Mac Build](#) [Source Code](#)

Companies & Projects Using Git
Google Microsoft Twitter LinkedIn NETFLIX PostgreSQL Android
RAILS Qt GNOME eclipse K X

[About this site](#)
Patches, suggestions, and comments are welcome.

Git is a member of [Software Freedom Conservancy](#)

2. 設定本地環境 打開終端機（Command Line）或 Git Bash，執行以下指令：

```
git config --global user.name "你的 GitHub 使用者名稱"  
git config --global user.email "你的 Email"
```

3. 創建新的儲存庫 (Repository)

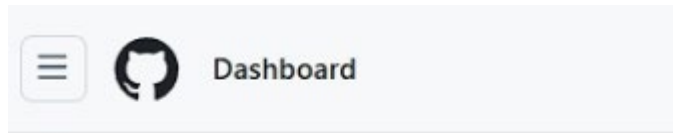
登入 GitHub，點擊「New repository」。

輸入專案名稱,將「Add README」改為 ON 讓空資夾也能下載檔案。

設定「儲存庫名稱」，選擇「Public（公開）」或「Private（私有）」。

點擊「Create repository」，完成儲存庫建立。

將網址複製,貼到記事本。



Top repositories



Find a repository...

1

General

Owner *

 s6351301-netizen

Repository name *

1150422

✓ 1150422 is available.

Great repository names are short and memorable. How about [fictional-memory?](#)

Description

0 / 350 characters

2

Configuration

Choose visibility *

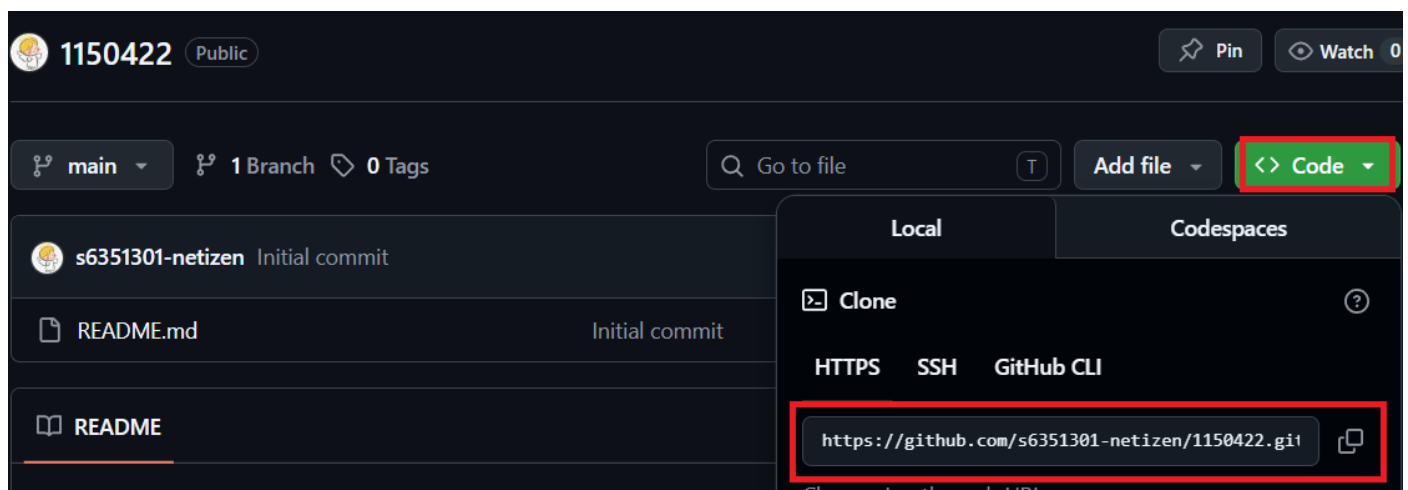
Choose who can see and commit to this repository

 Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

On ☒



4. 打開 Git Bash 軟體



5. 用 Git Bash 軟體下載 1150422 專案

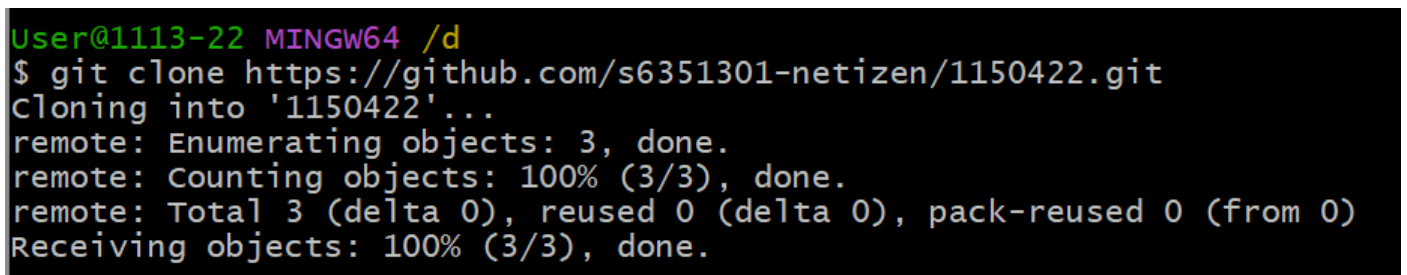
切到要下載的目錄 `cd d:/`

下載專案指令 `git clone` 網址

`$ git clone https://github.com/s6351301-netizen/1150422.git`

或將下載資料夾改名稱 `git clone` 網址 新資料夾名稱

`$ git clone https://github.com/s6351301-netizen/1150422.git may1150422`



要做檔案上傳或下載前,2 個先執行指令.

先把檔案加入 GIT 版控範圍 `git add`.

檔案加註說明 `git commit -m "說明文字"`

6. 檔案上傳 `git pull`

```
User@1113-22 MINGW64 /d
$ cd 1150422

User@1113-22 MINGW64 /d/1150422 (main)
$ ls
README.md

User@1113-22 MINGW64 /d/1150422 (main)
$ ls
README.md  index.html

User@1113-22 MINGW64 /d/1150422 (main)
$ git add .

User@1113-22 MINGW64 /d/1150422 (main)
$ git commit -m "新增index檔案"
[main dc33b17] 新增index檔案
1 file changed, 429 insertions(+)
create mode 100644 index.html

User@1113-22 MINGW64 /d/1150422 (main)
$ git pull
Already up to date.
```

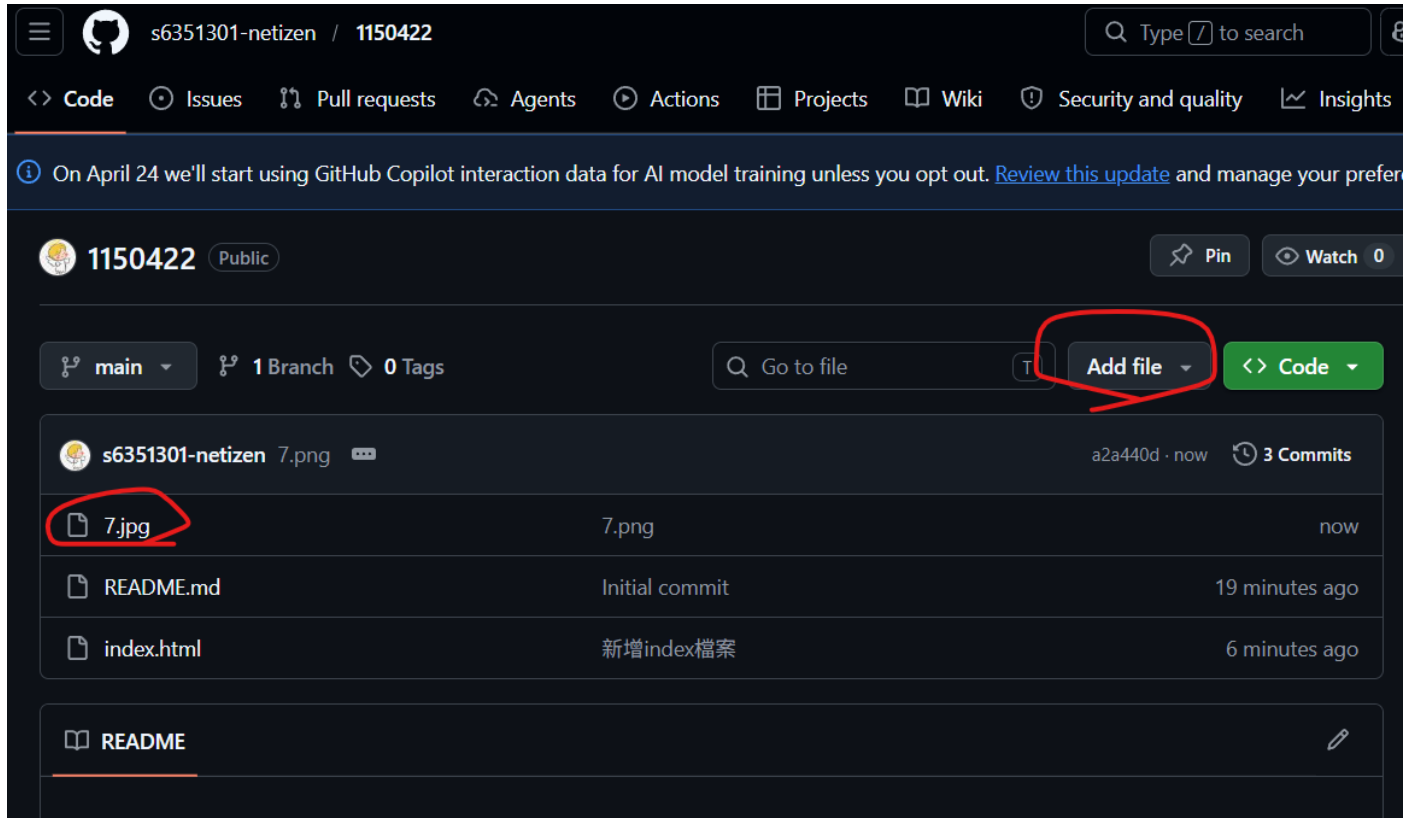
The screenshot shows the GitHub interface for the repository `s6351301-netizen / 1150422`. The repository is public and has 1 branch (main) and 0 tags. The commit history shows two commits:

File	Commit Message	Commit Hash	Time Ago
README.md	Initial commit	dc33b17	16 minutes ago
index.html	新增index檔案	dc33b17	3 minutes ago

The file `index.html` is circled in red in the original image. Below the commit history, there is a section for the README file.

7.檔案下載 git push

網頁上傳一張圖片



單純使用 `$ git push` 會出現因偉兩邊資料不同

需要強制更新 `$ git pull origin main`

就能正常下載新增 7 的圖片

```
User@1113-22 MINGW64 /d/1150422 (main)
$ git push
To https://github.com/s6351301-netizen/1150422.git
! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/s6351301-netizen/1150422.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

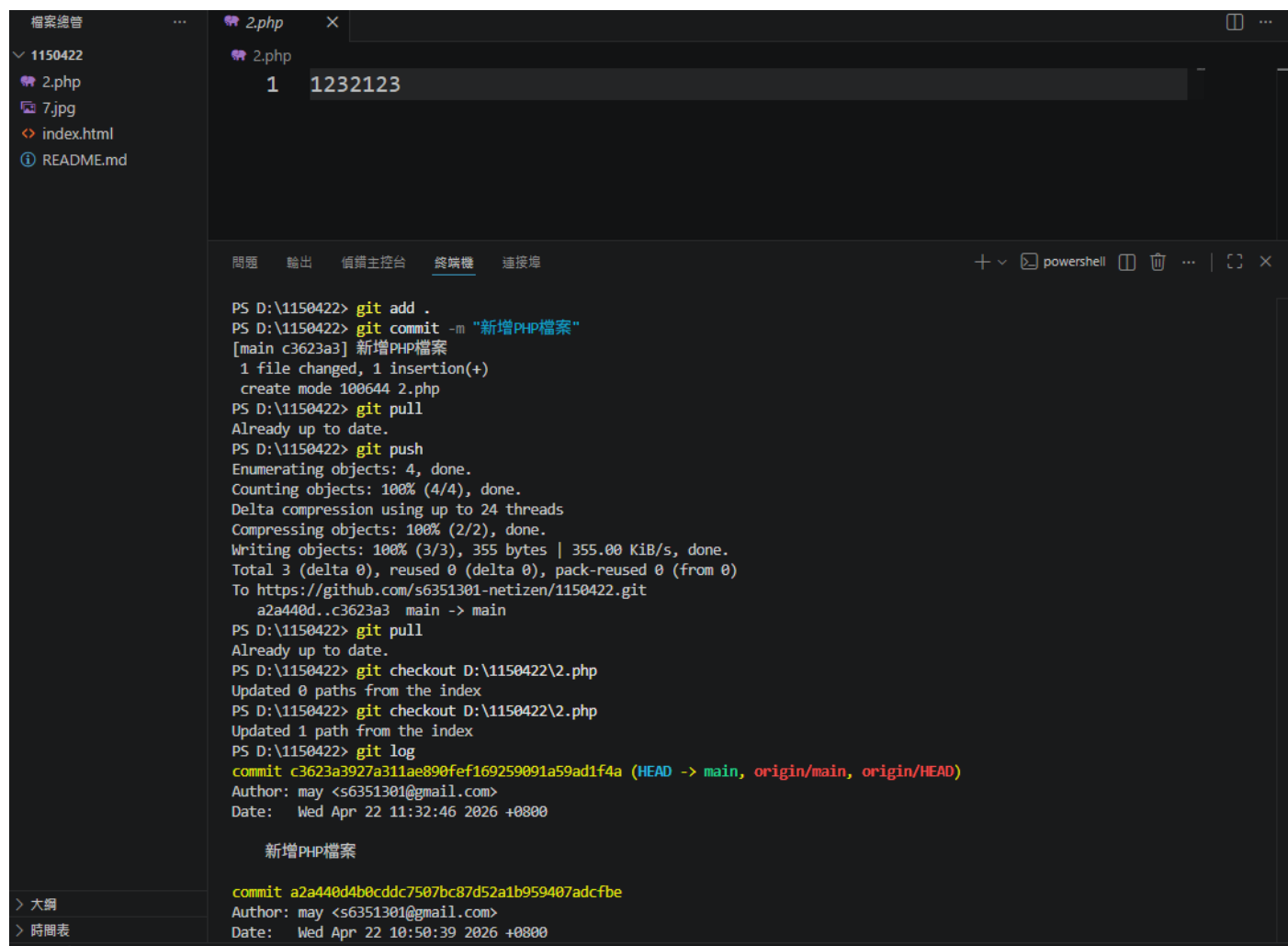
User@1113-22 MINGW64 /d/1150422 (main)
$ git pull origin main
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 12.62 KiB | 4.21 MiB/s, done.
From https://github.com/s6351301-netizen/1150422
* branch            main      -> FETCH_HEAD
   dc33b17..a2a440d  main      -> origin/main
Updating dc33b17..a2a440d
Fast-forward
 7.jpg | Bin 0 -> 12510 bytes
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 7.jpg
```

在 VS CORE 軟體 新增檔案並加入註解後上傳

git add .

git commit -m "新增 PHP 檔案"

git pull



```
PS D:\1150422> git add .
PS D:\1150422> git commit -m "新增PHP檔案"
[main c3623a3] 新增PHP檔案
 1 file changed, 1 insertion(+)
 create mode 100644 2.php
PS D:\1150422> git pull
Already up to date.
PS D:\1150422> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 24 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 355 bytes | 355.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/s6351301-netizen/1150422.git
 a2a440d..c3623a3  main -> main
PS D:\1150422> git pull
Already up to date.
PS D:\1150422> git checkout D:\1150422\2.php
Updated 0 paths from the index
PS D:\1150422> git checkout D:\1150422\2.php
Updated 1 path from the index
PS D:\1150422> git log
commit c3623a3927a311ae890fef169259091a59ad1f4a (HEAD -> main, origin/main, origin/HEAD)
Author: may <s6351301@gmail.com>
Date: Wed Apr 22 11:32:46 2026 +0800

    新增PHP檔案

commit a2a440d4b0cddc7507bc87d52a1b959407adcfbe
Author: may <s6351301@gmail.com>
Date: Wed Apr 22 10:50:39 2026 +0800
```

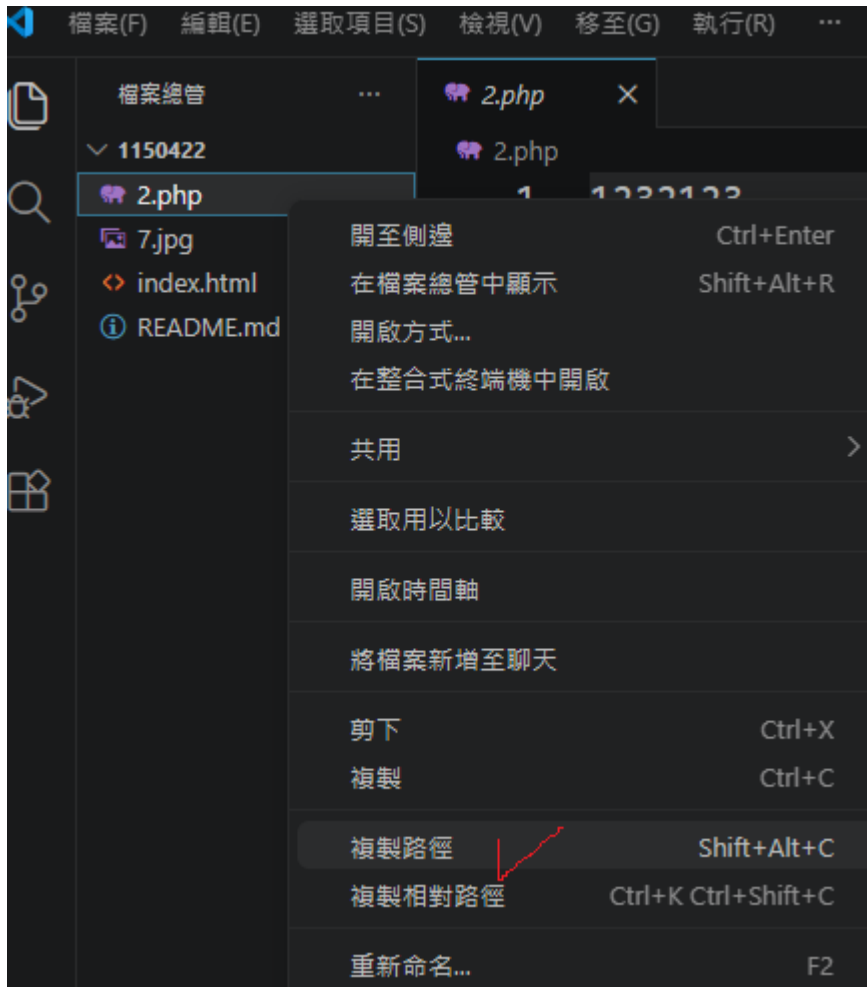

對新增檔案作內容新增 存檔後

下恢復上一步 複製路徑 將此路徑記憶

在下指令 `git checkout D:\1150422\2.php`

觀察是否檔案是否有恢復到上一步

也可先改之前做備份



查看下指令的歷史紀錄

```
PS D:\1150422> git log
commit c3623a3927a311ae890fef169259091a59ad1f4a (HEAD -> main, origin/main, origin/HEAD)
Author: may <s6351301@gmail.com>
Date:   Wed Apr 22 11:32:46 2026 +0800
```

新增PHP檔案

```
commit a2a440d4b0cddc7507bc87d52a1b959407adcfbe
Author: may <s6351301@gmail.com>
Date:   Wed Apr 22 10:50:39 2026 +0800
```

7.png

新增一張圖片

```
commit dc33b17d4d0e6e8e047df82f60bff39b71eab86c
Author: may <s6351301@gmail.com>
Date:   Wed Apr 22 10:44:12 2026 +0800
```

新增index檔案

```
commit 7f476dfa5e9107eaea316420b656cfe97c53e88d
Author: may <s6351301@gmail.com>
Date:   Wed Apr 22 10:31:10 2026 +0800
```

Initial commit